

Clase Contacto

```
public class Contacto {  
    private String name;  
    private String phoneNumber;  
    public Contacto(String name, String phoneNumber) {  
        this.name = name;  
        this.phoneNumber = phoneNumber;  
    }  
    public String getName() {  
        return name;  
    }  
    public String getPhoneNumber() {  
        return phoneNumber;  
    }  
    public static Contacto createContact(String name, String phoneNumber) {  
        return new Contacto(name, phoneNumber);  
    }  
    @Override  
    public boolean equals(Object obj) {  
        if (this == obj) {  
            return true;  
        }  
        if (obj == null || getClass() != obj.getClass()) {  
            return false;  
        }  
        Contacto contacto = (Contacto) obj;  
        return name.equalsIgnoreCase(contacto.name);  
    }  
}
```

Clase TelefonoMovil

```
import java.util.ArrayList;

public class TelefonoMovil {

    private String myNumber;

    private ArrayList<Contacto> myContacts;

    public TelefonoMovil(String myNumber) {

        this.myNumber = myNumber;

        myContacts = new ArrayList<>();

    }

    public boolean addNewContact(Contacto contact) {

        if (findContact(contact.getName()) >= 0) {

            return false;

        }

        myContacts.add(contact);

        return true;

    }

    public boolean updateContact(Contacto oldContact, Contacto newContact) {

        int foundPosition = findContact(oldContact);

        if (foundPosition < 0) {

            return false;

        }

        int existingContact = findContact(newContact.getName());

        if (existingContact >= 0 && existingContact != foundPosition) {

            return false;

        }

        myContacts.set(foundPosition, newContact);

        return true;

    }

    public boolean removeContact(Contacto contact) {
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    int foundPosition = findContact(contact);
    if (foundPosition < 0) {
        return false;
    }
    myContacts.remove(foundPosition);
    return true;
}

private int findContact(Contacto contact) {
    return myContacts.indexOf(contact);
}

private int findContact(String contactName) {
    for (int i = 0; i < myContacts.size(); i++) {
        Contacto contact = myContacts.get(i);
        if (contact.getName().equalsIgnoreCase(contactName)) {
            return i;
        }
    }
    return -1;
}

public Contacto queryContact(String name) {
    int position = findContact(name);
    if (position >= 0) {
        return myContacts.get(position);
    }
    return null;
}

public void printContacts() {
    System.out.println("Lista de contactos:");
    for (int i = 0; i < myContacts.size(); i++) {

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        System.out.println((i + 1) + ". "
            + myContacts.get(i).getName()
            + " -> "
            + myContacts.get(i).getPhoneNumber());
    }
}
}

```

Clase Main

```

import java.util.Scanner;

public class Main {

    private static Scanner scanner = new Scanner(System.in);
    private static TelefonoMovil telefono = new TelefonoMovil("600123456");

    public static void main(String[] args) {

        boolean salir = false;

        printActions();

        while (!salir) {

            System.out.println("\nElige una opción:");

            int opcion = Integer.parseInt(scanner.nextLine());

            switch (opcion) {

                case 0:

                    System.out.println("Saliendo...");

                    salir = true;

                    break;

                case 1:

                    telefono.printContacts();

                    break;

                case 2:

                    addNewContact();

                    break;
            }
        }
    }
}

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        case 3:
            updateContact();
            break;
        case 4:
            removeContact();
            break;
        case 5:
            queryContact();
            break;
        case 6:
            printActions();
            break;
        default:
            System.out.println("Opción no válida.");
    }
}

}

private static void printActions() {
    System.out.println("\nOpciones:");
    System.out.println("0 - Salir");
    System.out.println("1 - Imprimir contactos");
    System.out.println("2 - Añadir contacto");
    System.out.println("3 - Actualizar contacto");
    System.out.println("4 - Eliminar contacto");
    System.out.println("5 - Buscar contacto");
    System.out.println("6 - Mostrar menú");
}

private static void addNewContact() {
    System.out.println("Nombre:");

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String name = scanner.nextLine();
System.out.println("Número:");
String phone = scanner.nextLine();
Contacto contact = Contacto.createContact(name, phone);
if (telefono.addNewContact(contact)) {
    System.out.println("Contacto añadido correctamente.");
} else {
    System.out.println("El contacto ya existe.");
}
}

private static void updateContact() {
    System.out.println("Nombre del contacto a actualizar:");
    String oldName = scanner.nextLine();
    Contacto existingContact = telefono.queryContact(oldName);
    if (existingContact == null) {
        System.out.println("Contacto no encontrado.");
        return;
    }
    System.out.println("Nuevo nombre:");
    String newName = scanner.nextLine();
    System.out.println("Nuevo número:");
    String newPhone = scanner.nextLine();
    Contacto newContact = Contacto.createContact(newName, newPhone);
    if (telefono.updateContact(existingContact, newContact)) {
        System.out.println("Contacto actualizado correctamente.");
    } else {
        System.out.println("Error al actualizar el contacto.");
    }
}
}

```

```

private static void removeContact() {
    System.out.println("Nombre del contacto a eliminar:");
    String name = scanner.nextLine();
    Contacto existingContact = telefono.queryContact(name);
    if (existingContact == null) {
        System.out.println("Contacto no encontrado.");
        return;

        if (telefono.removeContact(existingContact)) {
            System.out.println("Contacto eliminado correctamente.");
        } else {
            System.out.println("No se pudo eliminar.");
        }
    }
}

private static void queryContact() {
    System.out.println("Nombre a buscar:");
    String name = scanner.nextLine();
    Contacto contact = telefono.queryContact(name);
    if (contact == null) {
        System.out.println("Contacto no encontrado.");
    } else {
        System.out.println("Nombre: " + contact.getName());
        System.out.println("Teléfono: " + contact.getPhoneNumber());
    }
}
}

```